

ally to 1.35. Most AMD motherboards will not automatically accept XMP settings for the RAM. So you'll have to manually input the timings and the voltage levels for your module. As with Intel, AMD also has power saving states akin to C-states that fall under AMD Cool'n'Quiet settings. Disable these and disable virtualisation (SVM) for better stability.

AMD has an excellent tool called Ryzen Master which helps you overclock right from within Windows.

OVERCLOCKING RAM

RAM modules come with pre-baked profiles on them, so if you have a good kit, you can easily select the right profile in your BIOS and you're good to go. Certain BIOS will not be able to automatically pull the XMP profiles from the RAM modules so you'll have to enter them manually. The numbers will be mentioned on a sticker on the modules. If not, you can use a software such as CPU-Z or SPDTool to read the different profiles stored on your RAM and then enter them manually into the BIOS. Most motherboards will also have a DRAM multiplier which you can change to increase the memory speed. The multiplier values like between 1.00x to 1.33x. If you change this multiplier, you will have to input the individual RAM timings as well. The average RAM module has 5 prominent timings and a lot more secondary timings. We'll stick with the primary timings for now. There's CL, tRCD, tRP, tRAS and CMD. Here's what they do:

- **CL:** CAS Latency. The time it takes between a command having been sent to the memory and when it begins to reply to it.
- **tRCD:** RAS to CAS Delay. The time it takes between the activation of the line (RAS) and the column (CAS) where the data are stored in the matrix.
- **tRP:** RAS Precharge. The time it takes between disabling the access to a line of data and the beginning of the access to another line of data.
- **tRAS:** Active to Precharge Delay. How long the memory has to wait until the next access to the memory can be initiated.
- **CMD:** Command Rate. The time

GOING EXTREME

Prepping your board

Extreme overclocking with liquid nitrogen will cause water vapour to condense on the cool components and water does not mix well with electronic components. In order to protect your motherboard, you will have to coat it with Vaseline or any other protective coating rated for use on electronics. You can even get industrial material such as liquid electrical tape which is easy to apply and remove. Any and all components that are going to be in close vicinity to the LN2 pot need to be insulated. Vaseline is best recommended because you can later it on and then heat the board using a hair dryer to ensure that the Vaseline creeps into all nooks and crannies. It takes the least effort and offers the best protection. Faber Castell art erasers are also commonly used for insulation but you have to be diligent to ensure proper coverage. Once vaseline or clay has been properly packed onto the entire CPU socket area, you can place additional layers of foam on the same area with a small cutout in the centre to allow the cooler to mate with the CPU. Additionally, you might want to keep a fan right next to the pot drawing the evaporating LN2 away. There's no hard and fast rule of how you should prep your motherboard for extreme overclocking. Just ensure that there's no room for condensation.



Delidding the CPU

To cool your CPU better, you can remove the integrated heatsink (IHS) that you CPU comes with and mate the cooler directly onto the CPU die. This process is extremely risky since you might damage the CPU during the delidding process. Different generations of CPUs use different methods to attach the IHS onto the CPU die. Some use thermal paste which is easier to remove while some use solder which is much more difficult to remove. Even if you do delid the CPU, you



are left with two choices, either replace the existing thermal paste with an aftermarket paste that is much more conductive or place the cooler directly onto the die. The silicon die is very fragile and you have to ensure that there's even pressure across the surface while mounting the heatsink or pot. If there's any unevenness, then the CPU die can chip off and you'll be left with a dead CPU. Lapping or polish the CPU surface does help with reducing CPU temperatures but the difference is minimal. Moreover, lapping or delidding will void your CPU warranty. With a delidded CPU you can drop temperatures by 4-6 degrees under load while direct die cooling will bring your temperatures down by an additional 1-2 degrees.

it takes between the memory chip having been activated and when the first command may be sent to the memory.

High-end motherboards will automatically calculate and alter these values based on your overclock. Lastly, if you RAM isn't stable after increasing the multiplier, you should increase the voltage levels. Always check the voltage levels after overclocking.

STABILITY TESTING

Once you've achieved the desired overclock. It's always best to stress test the CPU as that will tell you if the overclock is stable across all kinds of workloads. For this reason it is advisable to use multiple software and run them in a loop for a day or two. The end goal is to ensure that your system can run stable for multiple days after you've overclocked it. 